WSQ · COURSE SLIDES

Microsoft Azure Fundamentals (AZ-900)

TGS-2023036449

Cloud concepts · architecture · governance

Dr. Alfred Ang · v2

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UEN: 201200696W



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LEARNER ACCESS

Digital Attendance · TRAQOM

https://lms-tms.tertiaryinfotech.com/

Attendance is a compliance record

1. **Open the LMS attendance tile**
2. **Scan the TRAQOM QR code**
3. **Complete every required checkpoint**
4. **Confirm attendance status**

1 — Front checkpoint

Complete at the start of class.

2 — Assessment checkpoint

Complete before assessment begins.

3 — 75% minimum

Attendance requirement for assessment/funding.

4 — Evidence

Trainer confirms recorded status.

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General Trainer Profile



Trainer Name

Title / Designation

Role	_____
Qualifications	_____
Expertise	_____
Experience	_____
Contact	_____

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YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Founder & Principal Trainer

Role	Cloud, AI and automation educator
Qualifications	Doctorate · ACTA/ACLP-aligned training
Expertise	Azure · cloud architecture · governance
Experience	Technical course design and delivery
Contact	enquiry@tertiaryinfotech.com

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ADMIN · HOUSEKEEPING

Ground rules for a safe Azure lab

1 Subscription

Confirm the intended training subscription before every deployment.

SCOPE

2 Credentials

Use unique strong passwords; never share keys or paste secrets into screenshots.

SECRET

3 Cost

Use small SKUs and run cleanup immediately after verification.

BUDGET

4 Evidence

Capture only resource state required by the lab and assessment.

PROOF

THE POINT

If identity, subscription or cost scope is unclear, stop before creating resources.

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Learning outcomes and course evidence

1 LO1 · Assess

Compare solution boundaries, specifications, performance and security signals.

DECIDE

2 LO2 · Plan

Translate required controls into a staged Azure implementation plan.

PLAN

3 LO3 · Review

Use cost, health, activity and monitoring evidence to resolve issues.

REVIEW

4 Exam alignment

Cover all AZ-900 skills measured from 20 July 2026.

AZ-900

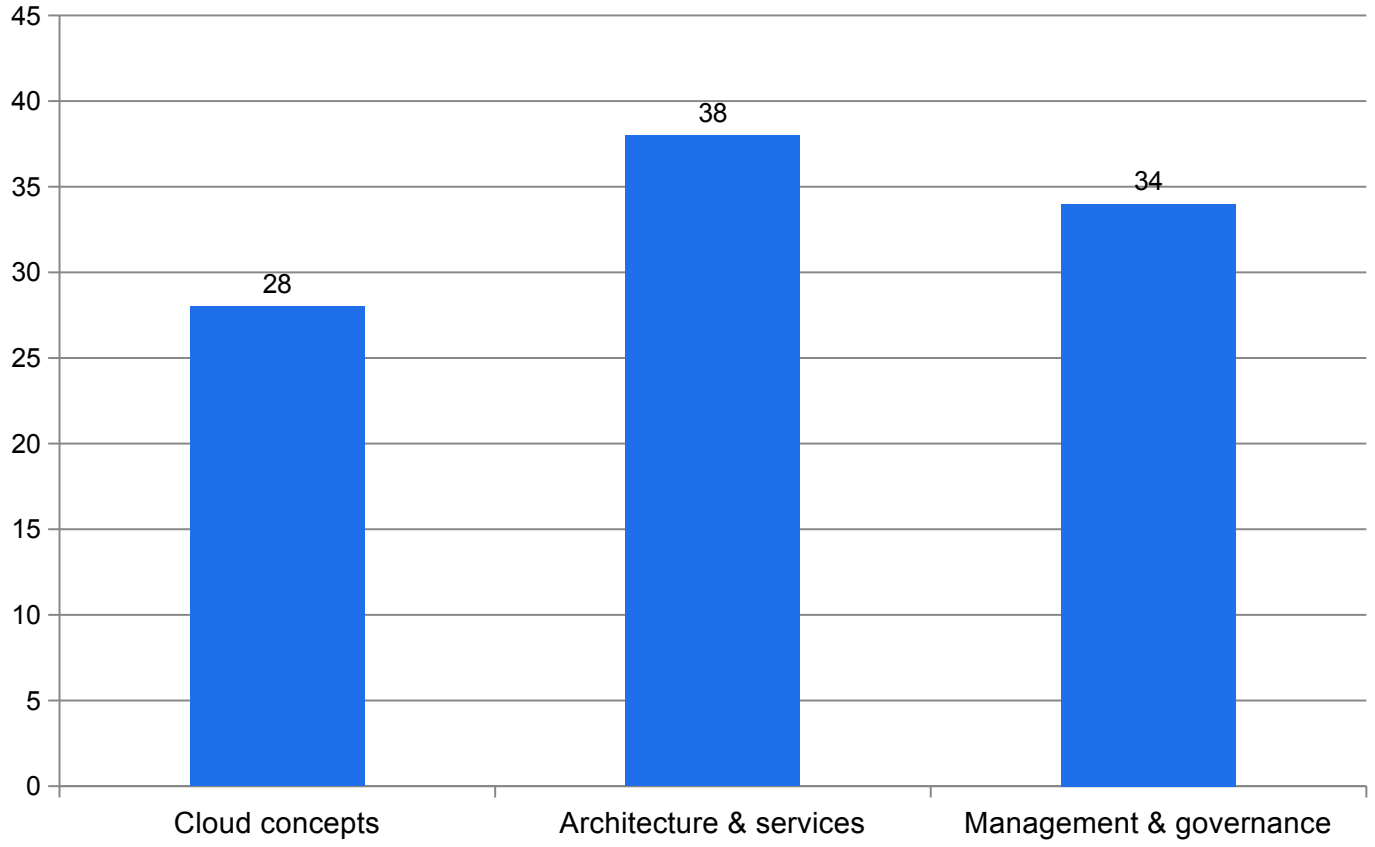
THE POINT

The labs create evidence for the WSQ outcomes; the concept slides explain the mechanisms behind that evidence.

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EDITABLE NATIVE CHART

AZ-900 domain weighting



WHAT THE DATA SHOWS

Midpoint teaching allocation from the official exam ranges: 25–30%, 35–40%, and 30–35%. Use the weighting to prioritise revision—not to skip any objective.

Midpoints of official exam-weight ranges

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Two-day lesson flow

1 Day 1 AM

Admin, cloud models, pricing, serverless, regions and control plane.

FOUNDATION

2 Day 1 PM

Compute, application hosting, containers and networking labs.

BUILD

3 Day 2 AM

Storage, data, identity, RBAC and governance mechanisms.

CONTROL

4 Day 2 PM

Cost, monitoring, cleanup, WA and PP assessment.

ASSESS

THE POINT

Detailed click paths and commands stay in the Learner Guide; the deck carries mechanisms, decisions and evidence.

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Briefing for assessment

1 Written Assessment

9 open-ended SAQs covering K1–K9; 1 hour; open book.

WA

2 Practical Performance

3 lab-aligned tasks covering A1–A5; 1 hour; open book.

PP

3 Submission

Complete the supplied documents and upload through the LMS.

LMS

4 Competence

All criteria require sufficient, valid and authentic evidence.

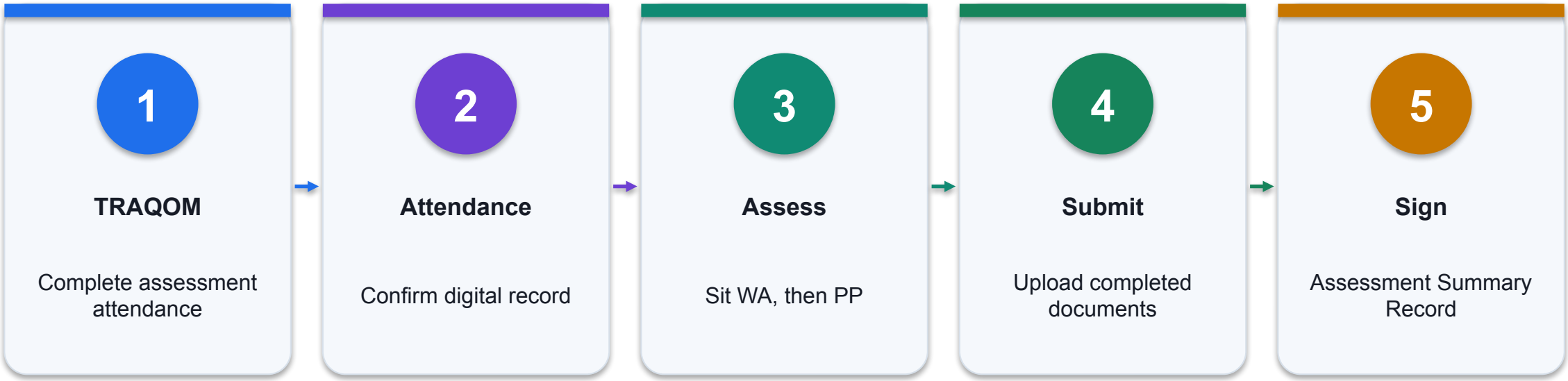
C / NYC

THE POINT

Question papers are learner-facing. Answer keys remain trainer-only and are never linked publicly.

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Assessment flow



CONTROL RULE

Use the LMS for attendance, assessment download and answer submission.

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LEARNER ACCESS

Download Course Material

https://lms-tms.tertiaryinfotech.com/

Courseware and assessment on the LMS

1. **Sign in with your learner account**
2. **Open this course**
3. **Download PPT/PDF and Learner Guide**
4. **Use the assessment tile only when instructed**

1 **Trainer slides**

PPTX is the editable trainer artifact.

2 **Learner slides**

PDF copy appears with the Learner Guide.

3 **Learner Guide**

Detailed steps, commands and checks.

4 **Assessment**

Upload completed candidate papers here.

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LEARNER ACCESS

Access the Hands-On Labs

github.com/tertiarycourses/TGS-2023036449-AZ-900

Two ways to get the public labs

1. **Clone the repository with Git**
2. **Or use Code → Download ZIP**
3. **Open labs/README.md**
4. **Follow the numbered labs in order**

1. **Public**
   Labs and learner courseware only.
2. **Private**
   Assessments and answer keys are excluded.
3. **Browser-first**
   Portal and Cloud Shell need no local install.
4. **Cleanup**
   Every billable lab ends with a cleanup gate.

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DOMAIN 1 · 25–30%

Describe Cloud Concepts

01

Cloud models · economics · benefits · service types

Labs 1–3 · Blueprint effective 20 July 2026

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TECHNICAL MODEL

Shared responsibility changes by service model

1 Customer always owns

Data, identities, accounts and access decisions remain customer responsibilities.

CONTROL

2 IaaS boundary

Customer manages OS, runtime and application; provider manages facilities and hardware.

IaaS

3 PaaS boundary

Provider manages OS and runtime; customer manages application configuration and data.

PaaS

4 SaaS boundary

Provider runs the application; customer governs users, data and permitted use.

SaaS

THE POINT

Ask who can change the control and who must supply the evidence—not merely who hosts the server.

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DECISION MATRIX

Public, private and hybrid cloud boundaries

Model	Ownership	Best-fit signal	Primary trade-off
Public	Provider-operated	Elastic demand; rapid service access	Less physical control
Private	Single organisation	Regulated or specialised infrastructure	Higher capital and skills burden
Hybrid	Connected public + private	Data residency or staged migration	Integration and governance complexity

A cloud model is an ownership and boundary decision; it is not a synonym for a product name.

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TECHNICAL MODEL

Consumption economics: translate usage into spend

1 Meter

Services measure units such as vCPU-hours, GB-months, transactions and outbound data.

INPUT

2 Rate

SKU, region, tier, reservation and commitment determine the price per unit.

RATE

3 Quantity

Scale, runtime, retention and traffic turn technical behaviour into billable usage.

USAGE

4 Control

Budgets, tags, autoscale and cleanup change future cost but do not erase consumed usage.

ACTION

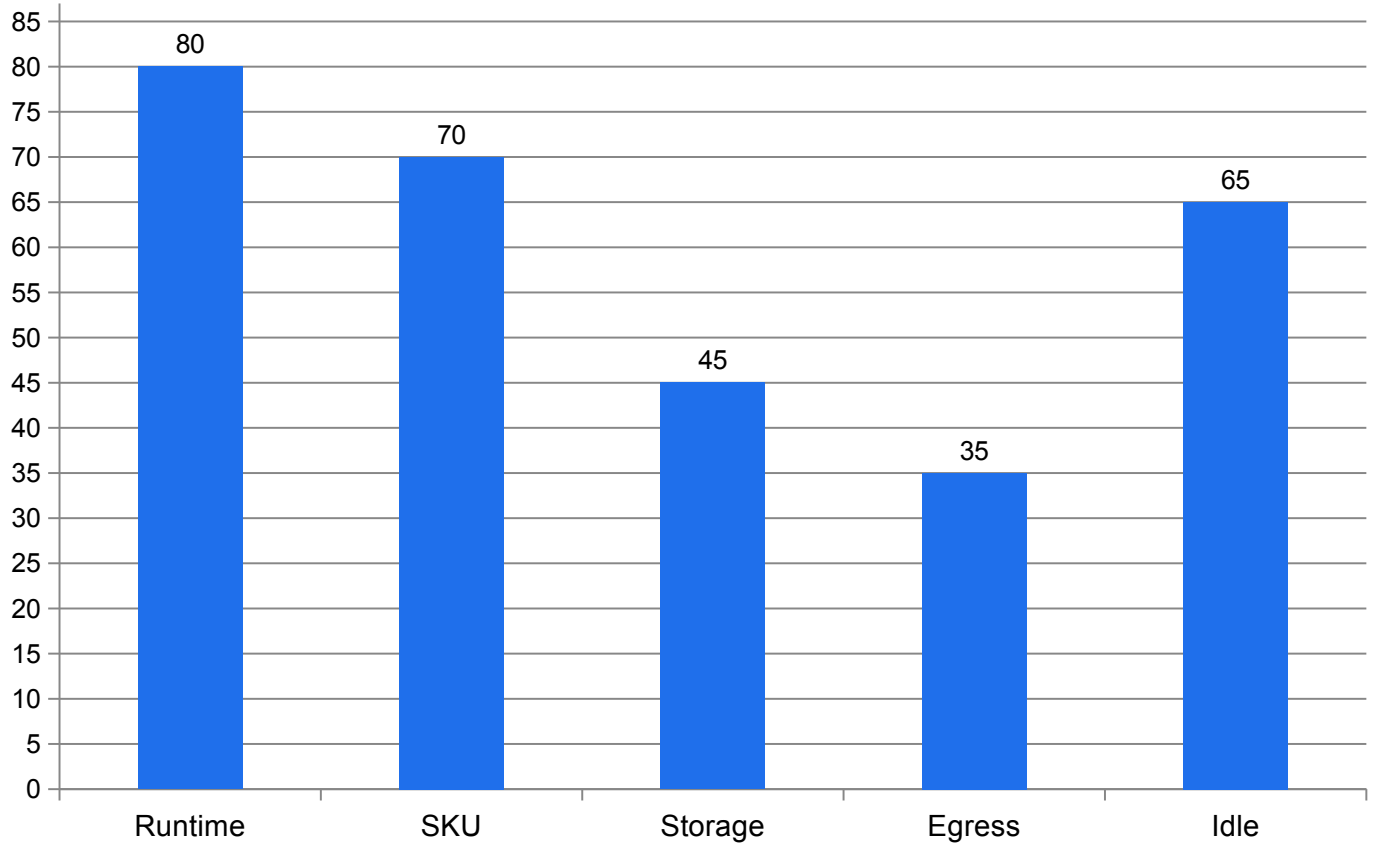
THE POINT

Estimated cost = metered quantity × applicable rate; validate both sides of the equation.

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EDITABLE NATIVE CHART

Illustrative cloud cost levers



WHAT THE DATA SHOWS

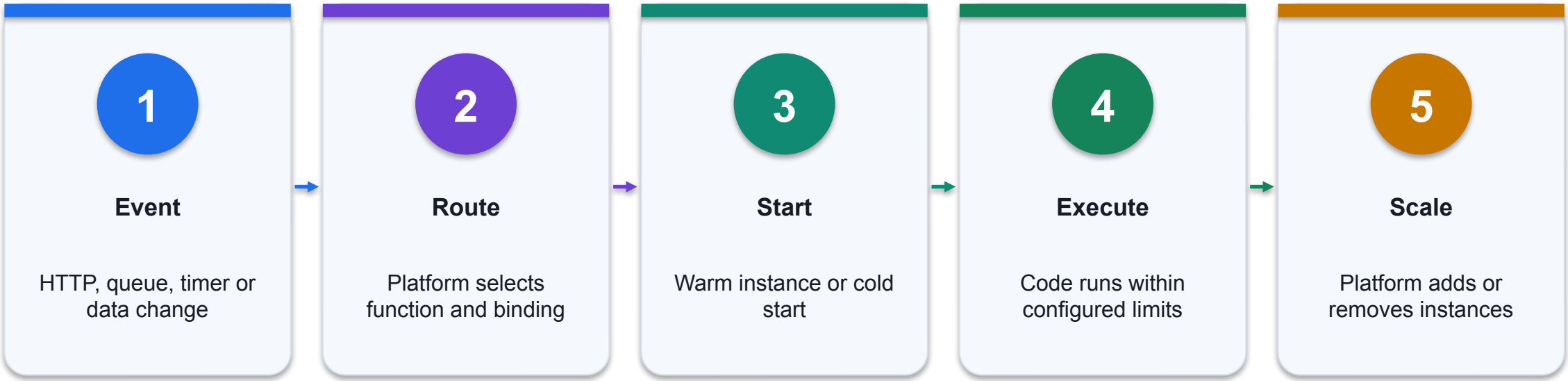
Indicative teaching scores: runtime, SKU choice and idle capacity usually dominate early lab spend. Actual charges depend on service, region and contract.

Midpoints of official exam-weight ranges

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MECHANISM TRACE

Serverless request lifecycle



CONTROL RULE

Serverless removes server management, not architecture, security, performance or cost decisions.

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Availability and scalability solve different problems

1 High availability

Redundant instances and zones reduce interruption when a component fails.

FAILURE

2 Vertical scale

Increase CPU, memory or IOPS of one instance; simple but bounded.

UP

3 Horizontal scale

Add instances and distribute load; resilient but requires stateless design.

OUT

4 Elasticity

Scale automatically with demand and return capacity when demand falls.

AUTO

THE POINT

Availability targets failure; scalability targets workload. A design normally needs both.

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TECHNICAL MODEL

Reliability and predictability need evidence

1 SLA

A service commitment expressed as availability over a defined period and scope.

COMMIT

2 Composite SLA

For serial dependencies, multiply decimal availability values to estimate the path SLA.

CALC

3 Performance

Capacity tests and monitoring reveal whether latency and throughput stay within targets.

MEASURE

4 Cost

Budgets and forecasts reduce surprise; they do not cap usage unless automation acts.

FORECAST

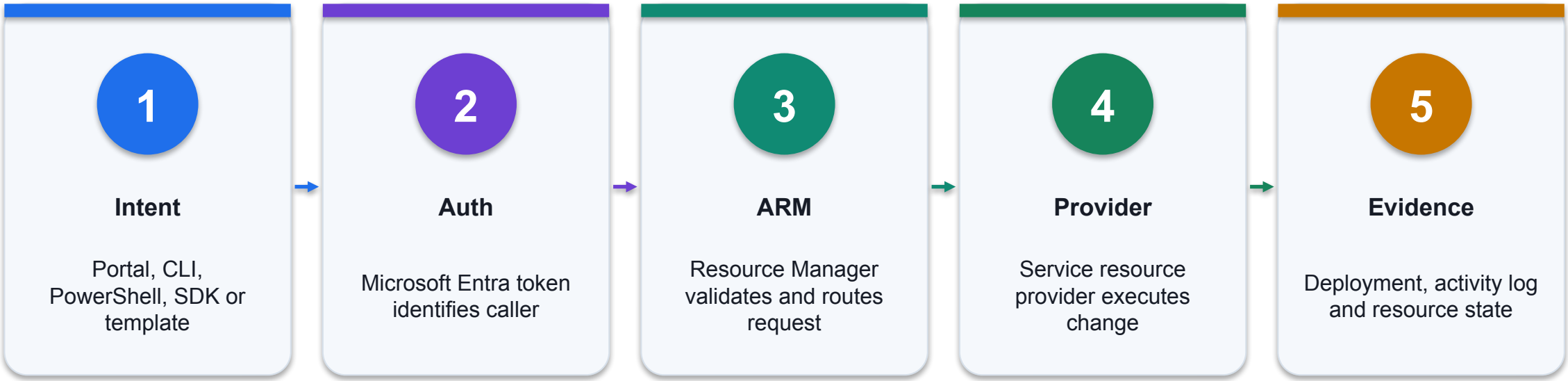
THE POINT

Predictability comes from explicit targets, measured telemetry and a response plan—not from the cloud label.

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MECHANISM TRACE

Manageability: one control plane, many interfaces



CONTROL RULE

Different tools converge on Azure Resource Manager; choose the interface that gives repeatability and evidence.

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DECISION MATRIX

IaaS, PaaS and SaaS decision matrix

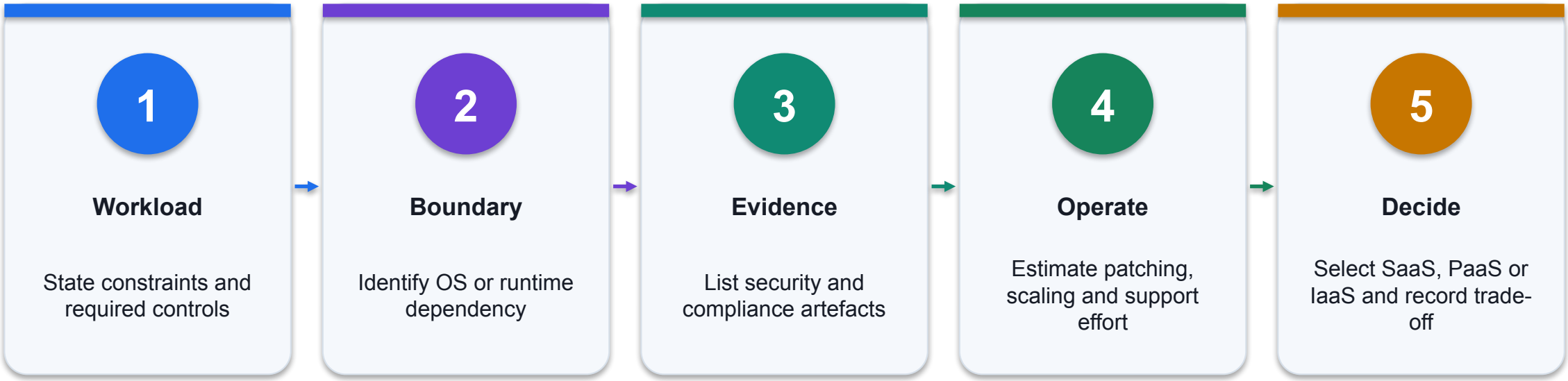
Type	You configure	Provider operates	Use when
IaaS	OS, runtime, app, data	Facilities, hardware, virtualisation	Legacy or OS-level control is required
PaaS	App, data, service configuration	OS, runtime, scaling primitives	Ship applications without server operations
SaaS	Users, data, tenant settings	Complete application stack	Adopt a finished business capability

Choose the highest abstraction that still exposes the controls your workload genuinely needs.

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MECHANISM TRACE

Service-model selection trace

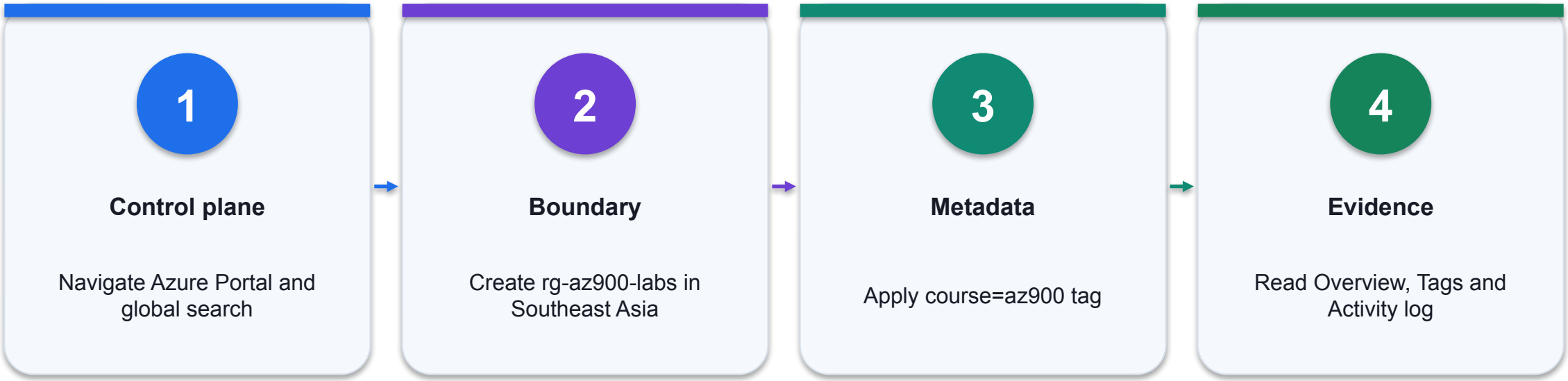


CONTROL RULE

Start from required control and evidence; do not begin with a favourite service.

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Lab 1 · Tour the Azure Portal and Create Your First Resource Group



CONTROL RULE

Deliverable: Resource group exists in Southeast Asia with course=az900 and no unexpected resources. Full click-by-click procedure: Learner Guide and labs/lab-01-*.md.

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Lab 1 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Resource group exists in Southeast Asia with course=az900 and no unexpected resources.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

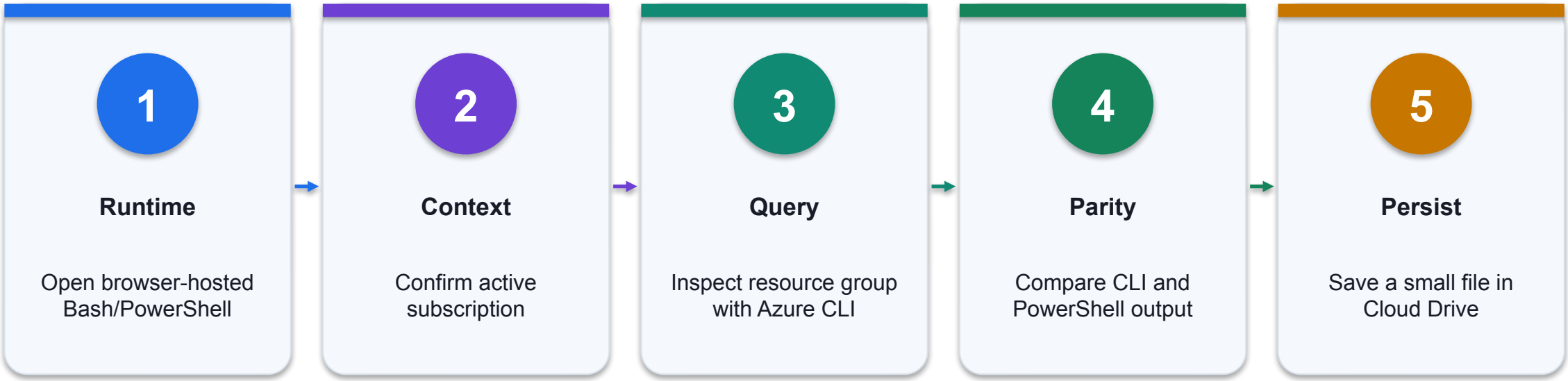
Wrong subscription or duplicate group name: confirm directory/subscription before continuing.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 2 · Get Started with Azure Cloud Shell



CONTROL RULE

Deliverable: CLI and PowerShell return the same subscription and rg-az900-labs state. Full click-by-click procedure: Learner Guide and labs/lab-02-*.md.

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Lab 2 · Verification and diagnosis

ACCEPTANCE EVIDENCE

CLI and PowerShell return the same subscription and rg-az900-labs state.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

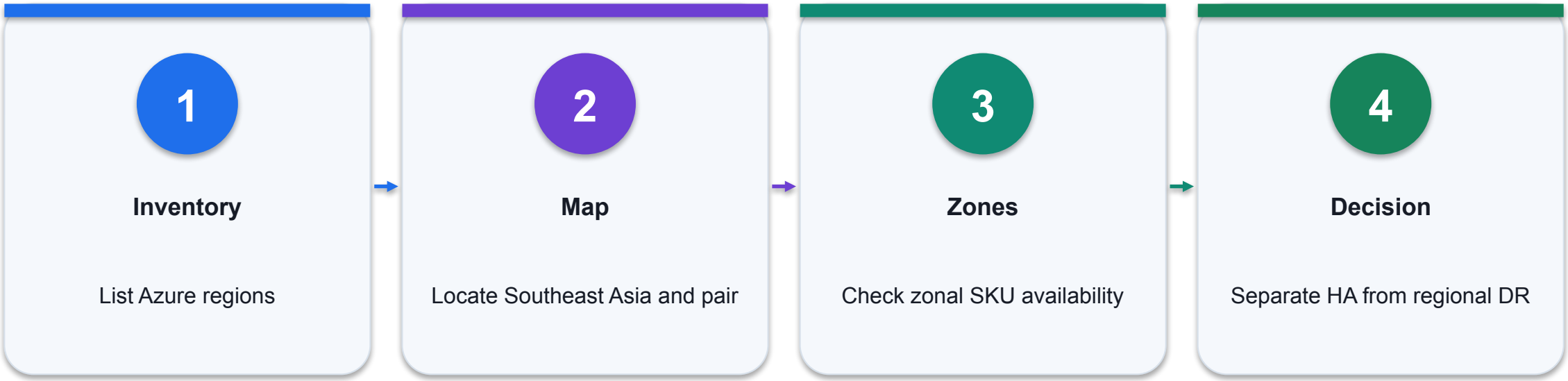
No prompt or wrong context: restart Cloud Shell and use az account set with the intended subscription.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 3 · Explore Regions, Region Pairs and Availability Zones



CONTROL RULE

Deliverable: Learner can name region, zone and region-pair boundaries and justify the lab region. Full click-by-click procedure: Learner Guide and labs/lab-03-*.md.

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Lab 3 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Learner can name region, zone and region-pair boundaries and justify the lab region.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

Assuming every service supports zones: verify capability for the selected service and region.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Domain 1 · Decision recap

1 Boundary

Responsibility changes as abstraction increases.

CONTROL

2 Economics

Usage and rate create spend.

COST

3 Resilience

HA, scale and recovery answer different failures.

DESIGN

4 Evidence

Targets, metrics and tests make promises verifiable.

VERIFY

THE POINT

Explain the mechanism, show the evidence, then state the trade-off.

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DOMAIN 2 · 35–40%

Describe Azure Architecture and Services

02

Global infrastructure · compute · networking · storage · identity

Labs 4–11 · Blueprint effective 20 July 2026

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VISUAL SYSTEM MODEL

Azure global fabric: region and zone boundaries



1 Region

A geographic market containing one or more datacenters connected by low-latency networking.

2 Availability zone

A physically separate datacenter location with independent power, cooling and networking.

3 Region pair

A platform planning relationship for recovery sequencing; not automatic application failover.

Use zones for datacenter-level resilience and a multi-region design for regional disaster recovery.

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DECISION MATRIX

Regions, zones and region pairs

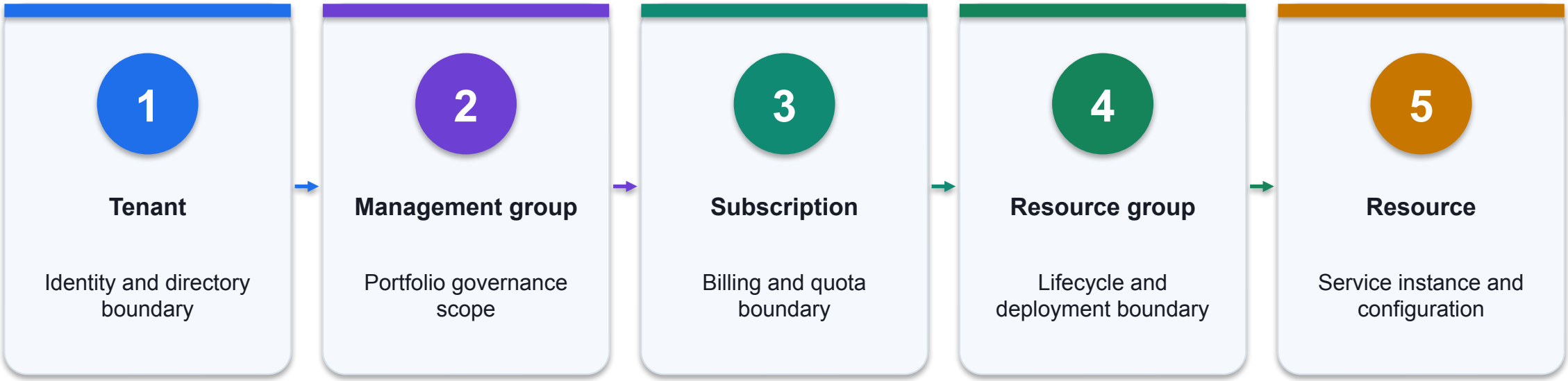
Boundary	Protects against	Design action	Evidence
Region	Geographic/data residency constraint	Select supported service and latency location	Resource location
Availability zone	Datacenter failure	Place redundant zonal or zone-redundant instances	Zone field / architecture
Region pair	Regional event planning	Design replication and tested failover separately	Recovery plan

A region pair is not a substitute for configuring replication, backups or failover.

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MECHANISM TRACE

Azure resource hierarchy and inheritance



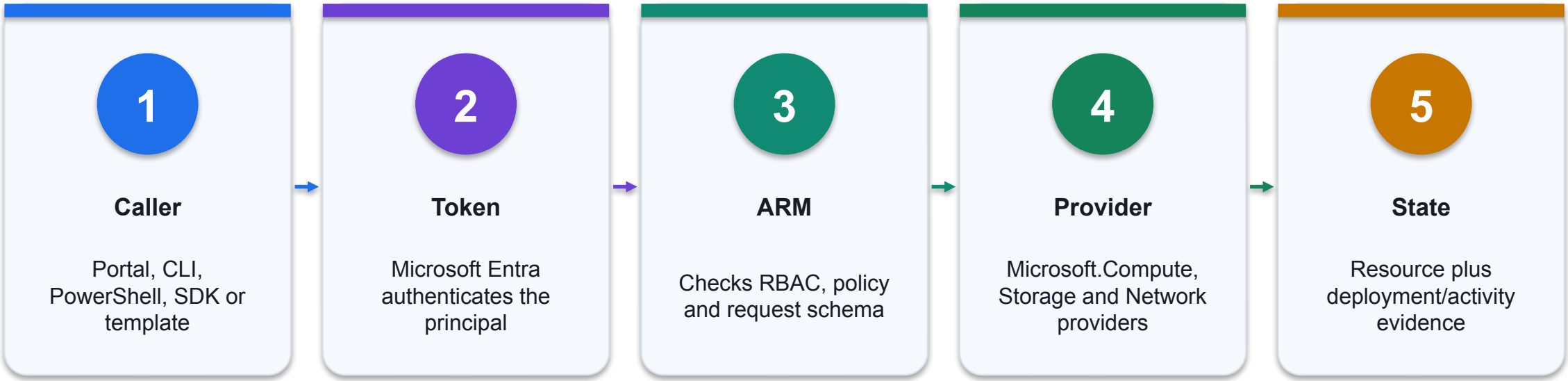
CONTROL RULE

RBAC and Policy can inherit downward; resource groups cannot be nested.

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MECHANISM TRACE

Azure Resource Manager request path



CONTROL RULE

The control plane changes resource configuration; the data plane moves or reads workload data.

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DECISION MATRIX

Compute selection: VM, containers or functions

Option	Runtime control	Scale unit	Best-fit signal
Virtual machine	Full guest OS	VM / scale-set instance	OS dependency or lift-and-shift
Container	Image and process	Container group / replica	Portable packaged service
Function	Function code + bindings	Invocation / instance	Event-driven, short-lived work

Choose by required control, state model, event pattern and operations burden—not by novelty.

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TECHNICAL MODEL

Virtual machine resource contract

1 Compute

Size defines vCPU, memory, temporary disk and throughput ceilings.

SKU

2 Image

Marketplace or custom image provides OS and starting configuration.

OS

3 Storage

Managed OS/data disks persist independently of temporary disk.

DISK

4 Network

NIC, subnet, NSG and optional public IP define reachability.

NET

THE POINT

A VM is a graph of resources; deleting only the VM can leave disks, NICs and public IPs billable.

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DECISION MATRIX

VM availability choices

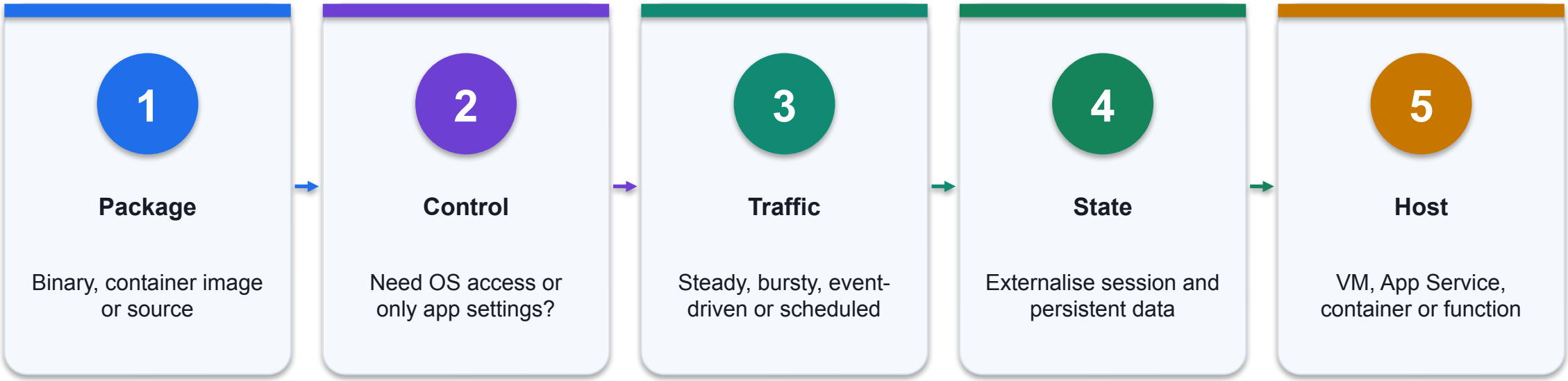
Mechanism	Failure scope	Placement	Use
Availability set	Rack/update domain	Logical distribution in one datacenter	Legacy regional VM pattern
Availability zones	Datacenter	Explicit zones in a region	Higher regional resilience
VM Scale Sets	Instance failure + demand	Uniform/flexible fleet	Autoscaling identical workloads
Azure Virtual Desktop	User session workload	Pooled/personal desktops	Managed virtual desktop delivery

Availability is an application property: add load distribution, health probes and data resilience.

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MECHANISM TRACE

Application hosting decision



CONTROL RULE

Match the hosting abstraction to runtime control and workload behaviour.

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Container runtime contract

1 Image

Immutable layers define filesystem and startup command.

BUILD

2 Registry

Stores tagged image versions and access policy.

PULL

3 Instance

Runs the image with CPU, memory, network and environment settings.

RUN

4 Observability

Logs, restart state and health evidence explain runtime outcome.

VERIFY

THE POINT

A container is not a tiny VM: it is an isolated process that shares the host kernel.

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TECHNICAL MODEL

Virtual network building blocks

1 Address space

CIDR range defines private address capacity and overlap risk.

CIDR

2 Subnet

Segments workloads and becomes a scope for routes, NSGs and service integration.

SEGMENT

3 Peering

Connects VNets over Azure backbone without transitive routing by default.

CONNECT

4 DNS

Resolves names; private zones support private endpoint and internal service discovery.

RESOLVE

THE POINT

Plan non-overlapping CIDR space before deployment; renumbering connected networks is expensive.

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DECISION MATRIX

Public and private endpoints

Endpoint	Address	Traffic path	Control
Public	Internet-routable	Public network to service edge	Firewall, auth, TLS, allow lists
Private	Private IP in VNet	Private link to service	Subnet, DNS, NSG plus service policy

A private endpoint changes the network path; it does not replace identity, authorisation or encryption.

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DECISION MATRIX

Hybrid connectivity options

Option	Transport	Typical use	Trade-off
VPN Gateway	Encrypted tunnel over internet	Fast hybrid connection and backup	Variable internet latency
ExpressRoute	Private provider circuit	Predictable enterprise connectivity	Lead time and circuit cost
VNet peering	Azure backbone	Connect Azure VNets	Not transitive by default
Azure DNS	Authoritative name service	Public/private name resolution	Needs correct zone linking

Connectivity design must include routing, DNS, security and failure behaviour—not just a tunnel.

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DECISION MATRIX

Azure Storage service selection

Service	Data model	Protocol/API	Use
Blob	Objects	HTTP/REST	Documents, media, backups, data lakes
Files	Managed file share	SMB/NFS	Lift-and-shift shared files
Queue	Messages	REST	Loose coupling and background work
Table	NoSQL key/attribute	REST	Massive simple structured datasets

Select storage by access pattern, protocol, consistency and lifecycle—not by file extension.

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TECHNICAL MODEL

Blob access tiers and lifecycle

1 Hot

Higher storage price, lower access cost; frequent reads and writes.

FREQUENT

2 Cool

Lower storage price with minimum retention and higher access cost.

INFREQUENT

3 Cold

Lower cost for seldom-accessed online data with longer minimum retention.

RARE

4 Archive

Offline tier requiring rehydration before read; lowest storage cost.

OFFLINE

THE POINT

Lifecycle rules can move blobs by age; retention, retrieval time and access charges still apply.

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DECISION MATRIX

Storage redundancy contract

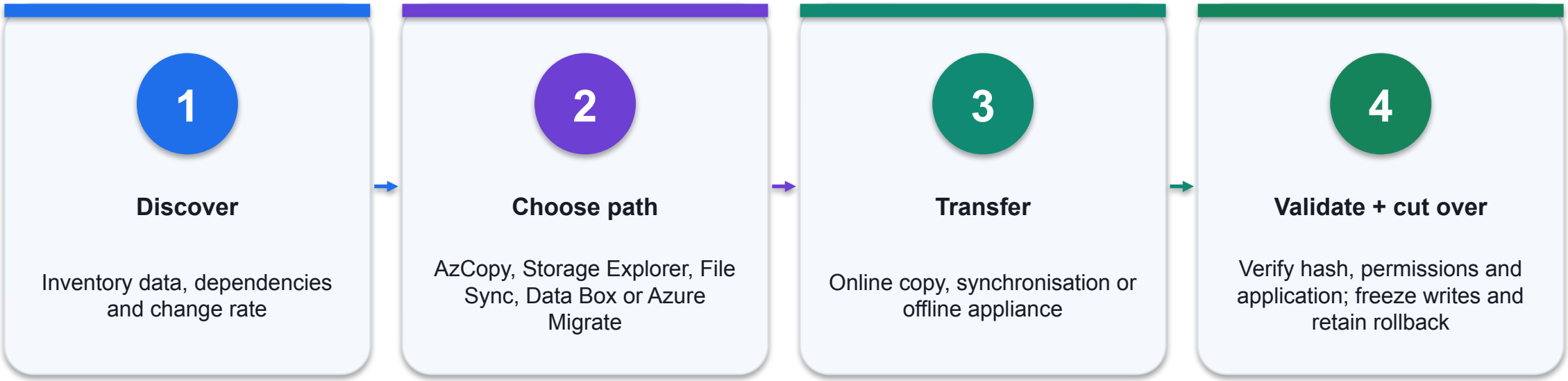
Option	Copies	Failure scope	Read secondary
LRS	3 in one datacenter	Disk/rack	No
ZRS	Across 3 zones	Datacenter	No
GRS	Primary LRS + paired region LRS	Region after failover	No
RA-GRS	GRS plus secondary read endpoint	Region + read continuity	Yes

Redundancy protects service data copies; it does not replace backup, versioning or deletion protection.

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MECHANISM TRACE

Move and migrate data



CONTROL RULE

A migration is complete only when data, permissions, application behaviour and rollback evidence are verified.

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TECHNICAL MODEL

Microsoft Entra directory boundary

1 Identity

Users, groups, service principals and managed identities are security principals.

WHO

2 Authentication

Passwords, MFA, passwordless and federation prove identity.

PROVE

3 Conditional Access

Evaluates signal and policy before token issuance.

DECIDE

4 Authorisation

Azure RBAC and application roles decide permitted actions.

ALLOW

THE POINT

Authentication answers who you are; authorisation answers what you may do at a scope.

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DECISION MATRIX

Authentication method trade-offs

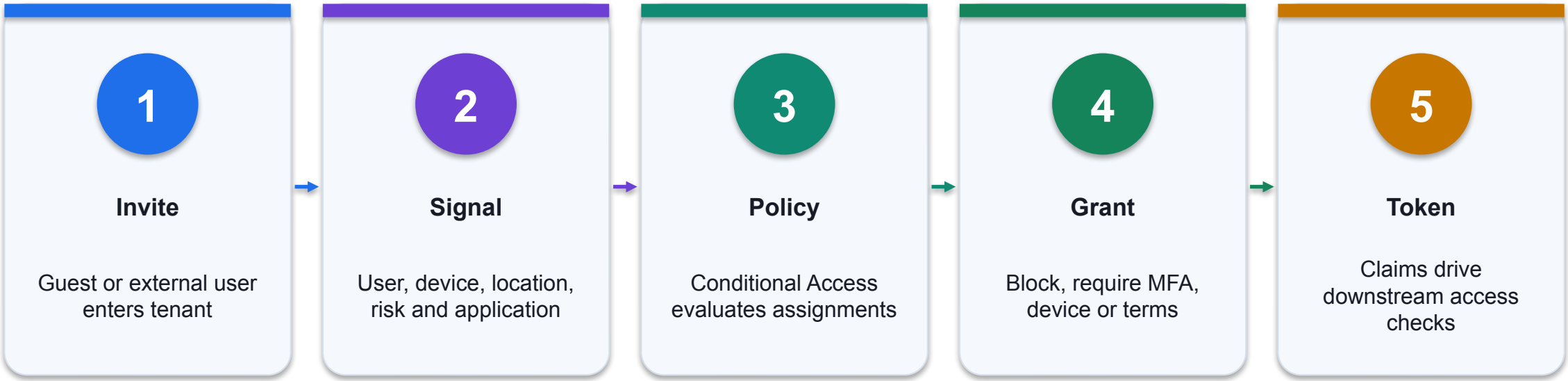
Method	User experience	Phishing resistance	Control
Password only	Familiar	Low	Password policy and risk monitoring
MFA	Second proof	Medium; method-dependent	Conditional Access requirement
Passwordless	Key/biometric/device	High with FIDO2/passkeys	Registration and device assurance
SSO	One sign-in across apps	Depends on primary method	Federation/token lifetime

SSO reduces repeated sign-in; it is not itself a strong authentication method.

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MECHANISM TRACE

External identity and Conditional Access trace



CONTROL RULE

External collaboration needs lifecycle, access review and sign-in policy—not just an invitation.

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TECHNICAL MODEL

RBAC assignment is a three-part tuple

1 Principal

User, group, service principal or managed identity receives access.

WHO

2 Role

A role definition lists allowed and excluded management/data actions.

WHAT

3 Scope

Management group, subscription, resource group or resource bounds access.

WHERE

4 Evidence

Role assignments and Check access show direct and inherited permissions.

VERIFY

THE POINT

Apply least privilege at the narrowest practical scope and prefer groups over repeated user assignments.

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TECHNICAL MODEL

Zero Trust and defense in depth

1 Verify explicitly

Use identity, device, risk and resource signals for each access decision.

IDENTITY

2 Least privilege

Minimise permissions, duration and reachable scope.

ACCESS

3 Assume breach

Segment networks, monitor activity and test recovery.

DESIGN

4 Layers

Physical, identity, perimeter, network, compute, application and data controls overlap.

DEPTH

THE POINT

No single control is the boundary; layered controls limit blast radius when one control fails.

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MECHANISM TRACE

Microsoft Defender for Cloud control loop

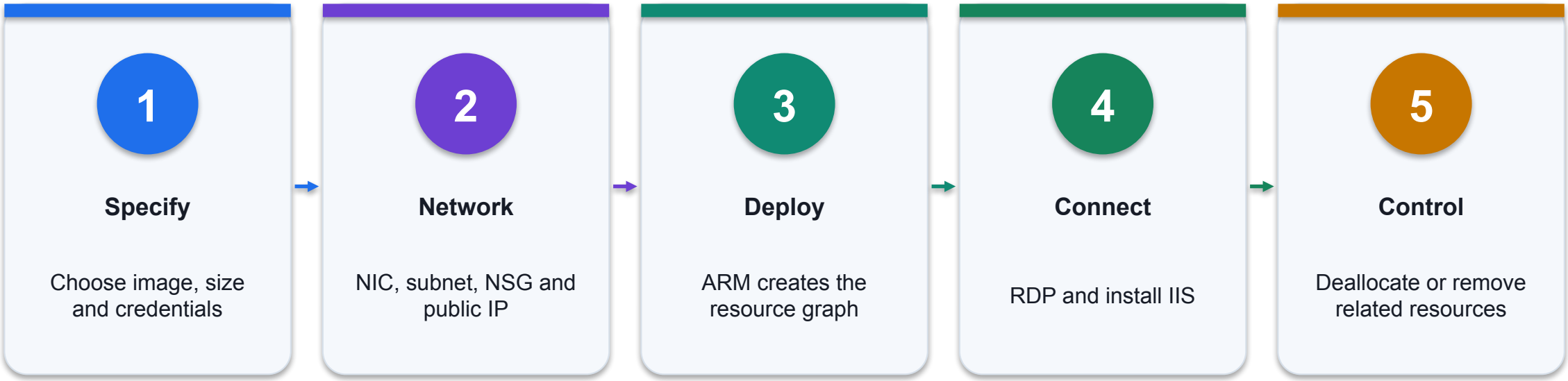


CONTROL RULE

Defender for Cloud combines posture management and workload protection; scope and plan settings matter.

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Lab 4 · Create a Windows Virtual Machine in the Portal



CONTROL RULE

Deliverable: VM overview is healthy and the IIS page is reachable only through the intended rule. Full click-by-click procedure: Learner Guide and labs/lab-04-*.md.

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Lab 4 · Verification and diagnosis

ACCEPTANCE EVIDENCE

VM overview is healthy and the IIS page is reachable only through the intended rule.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

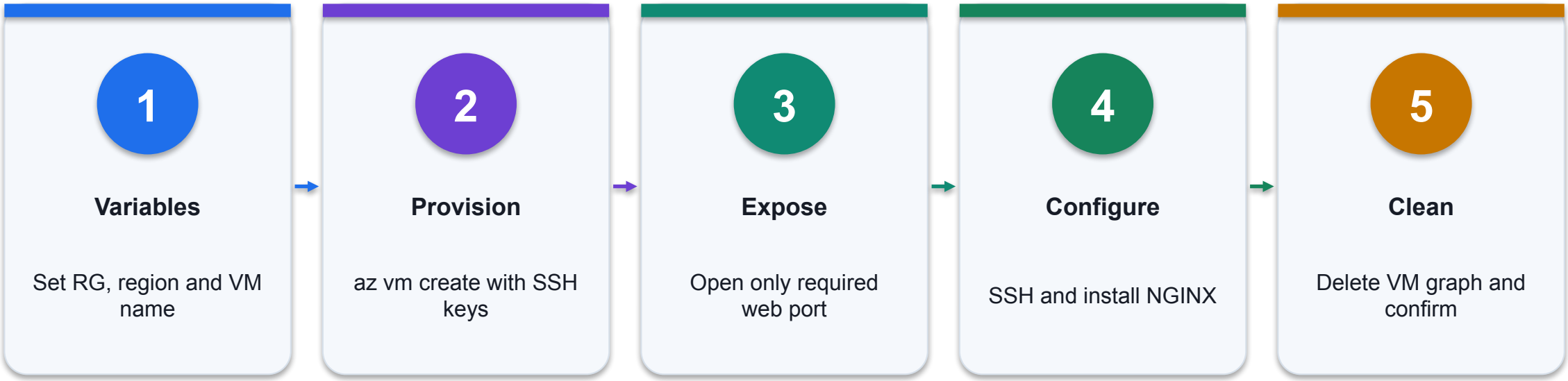
Timeout: check power state, public IP, NSG inbound rule and guest firewall.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 5 · Create a Linux VM with Azure CLI



CONTROL RULE

Deliverable: az vm get-instance-view reports running before the web test; cleanup leaves no attached VM resources. Full click-by-click procedure: Learner Guide and labs/lab-05-*.md.

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Lab 5 · Verification and diagnosis

ACCEPTANCE EVIDENCE

az vm get-instance-view reports running before the web test; cleanup leaves no attached VM resources.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

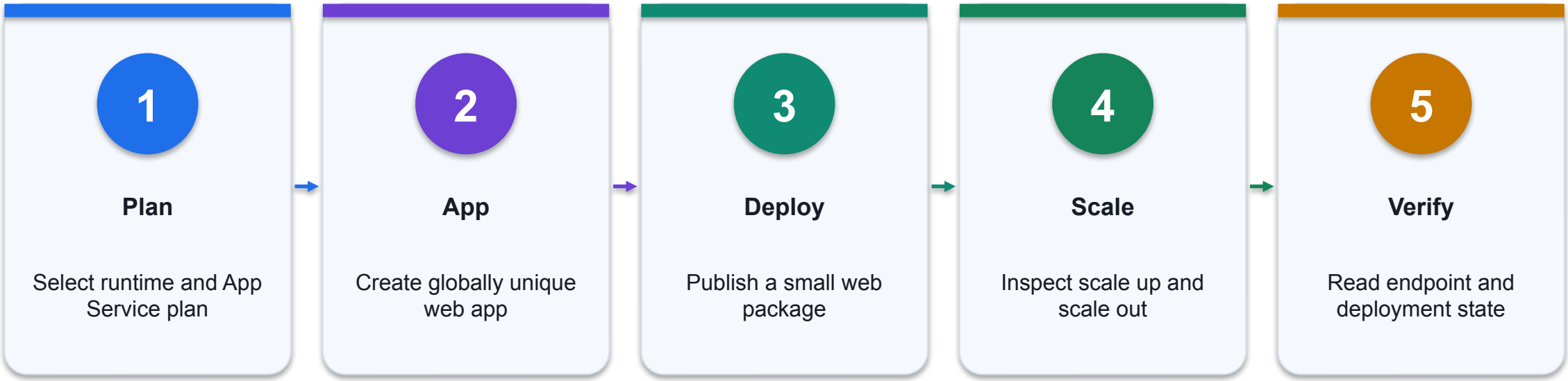
Image/size unavailable: query valid images and SKUs in Southeast Asia before retrying.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 6 · Create a Web App with Azure App Service



CONTROL RULE

Deliverable: HTTPS endpoint serves the deployed page and the deployment log shows success. Full click-by-click procedure: Learner Guide and labs/lab-06-*.md.

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Lab 6 · Verification and diagnosis

ACCEPTANCE EVIDENCE

HTTPS endpoint serves the deployed page and the deployment log shows success.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

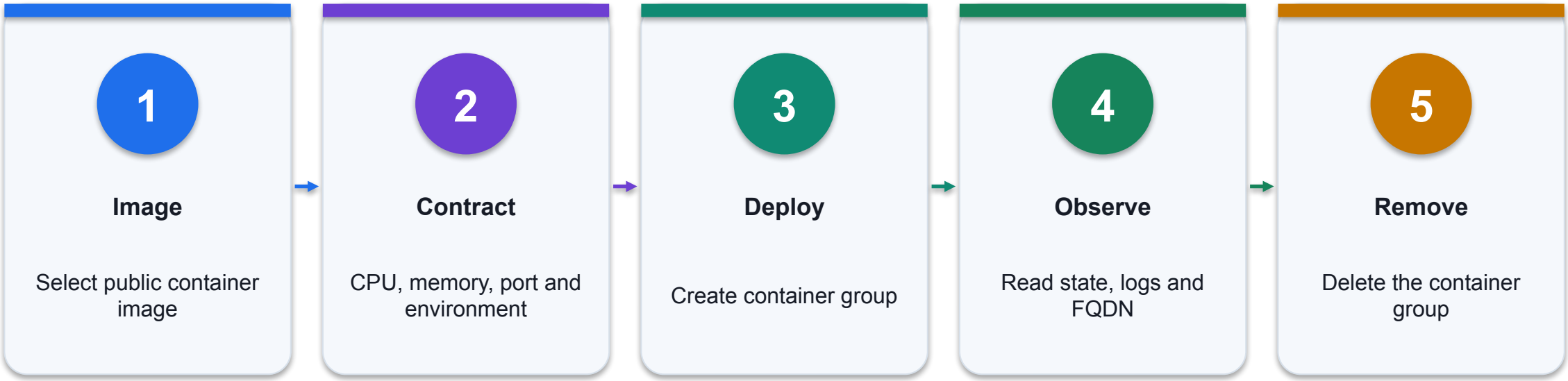
Default page persists: check deployment target, startup/runtime setting and deployment log.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 7 · Deploy a Container with Azure Container Instances



CONTROL RULE

Deliverable: Container state is Running, logs are readable and the FQDN returns expected content. Full click-by-click procedure: Learner Guide and labs/lab-07-*.md.

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Lab 7 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Container state is Running, logs are readable and the FQDN returns expected content.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

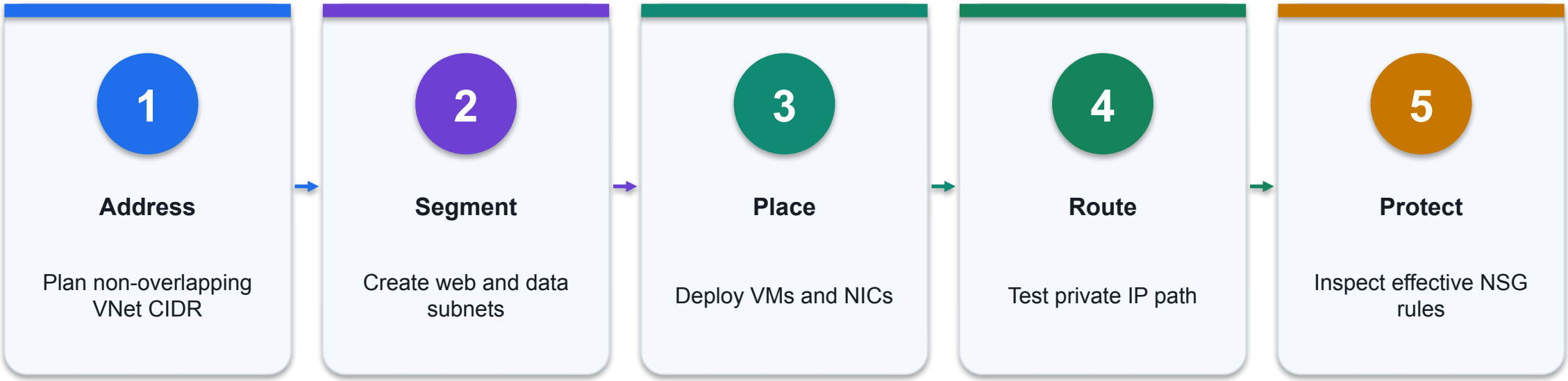
ImagePull or restart loop: inspect events, registry access, command and container logs.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 8 · Create a Virtual Network and Test Private Connectivity



CONTROL RULE

Deliverable: Web VM reaches data VM over private IP while the data VM has no public IP. Full click-by-click procedure: Learner Guide and labs/lab-08-*.md.

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Lab 8 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Web VM reaches data VM over private IP while the data VM has no public IP.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

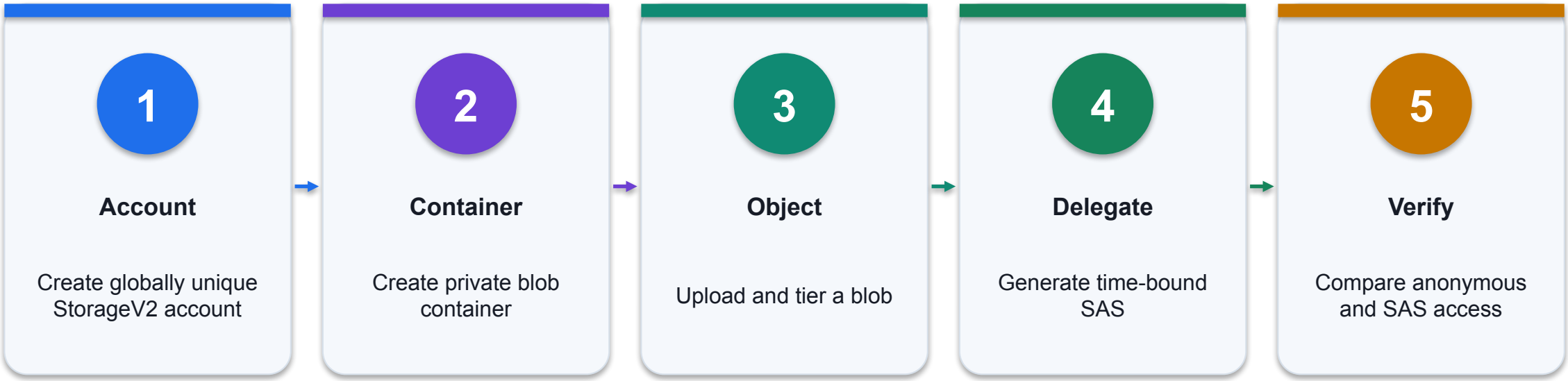
Private test fails: check NIC IP, subnet route, NSG rules and guest firewall.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 9 · Create a Storage Account and Work with Blob Storage



CONTROL RULE

Deliverable: Anonymous access fails, SAS access succeeds within its scope and the blob tier reads back correctly. Full click-by-click procedure: Learner Guide and labs/lab-09-*.md.

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Lab 9 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Anonymous access fails, SAS access succeeds within its scope and the blob tier reads back correctly.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

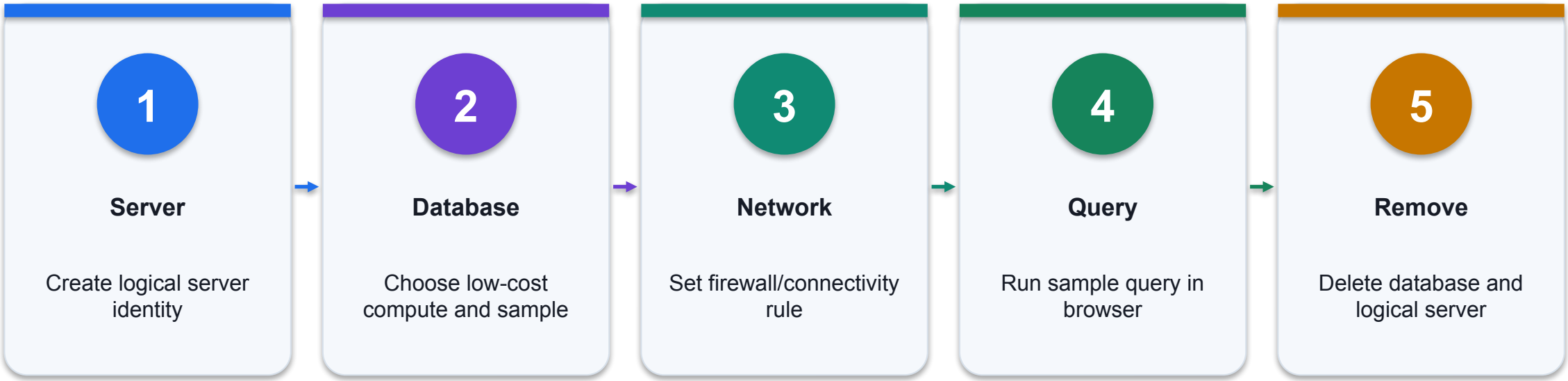
SAS fails: check expiry, permissions, resource type, clock and copied query string.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 10 · Create an Azure SQL Database



CONTROL RULE

Deliverable: Query returns rows from the sample database and firewall scope matches the intended client. Full click-by-click procedure: Learner Guide and labs/lab-10-*.md.

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Lab 10 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Query returns rows from the sample database and firewall scope matches the intended client.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

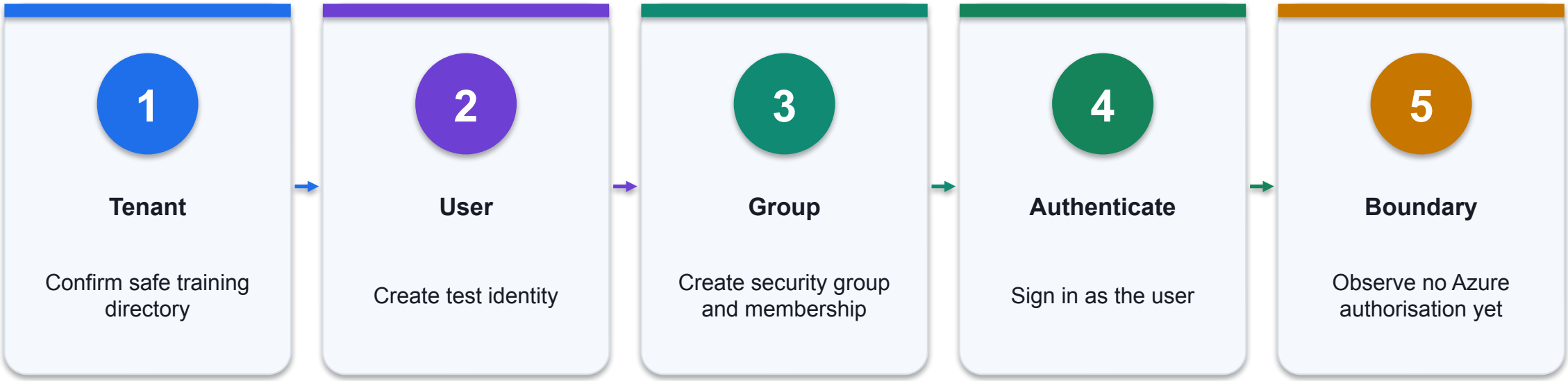
Login/query blocked: check server name, admin identity, firewall rule and database selection.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 11 · Explore Microsoft Entra ID Users and Groups



CONTROL RULE

Deliverable: Test user can authenticate but cannot manage subscription resources without RBAC assignment. Full click-by-click procedure: Learner Guide and labs/lab-11-*.md.

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Lab 11 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Test user can authenticate but cannot manage subscription resources without RBAC assignment.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

Using a corporate tenant: stop and switch to an authorised training tenant before creating identities.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Domain 2 · Decision recap

1 Hierarchy

Tenant to resource sets scope and inheritance.

SCOPE

2 Services

Select compute, network and storage by contract.

CHOOSE

3 Identity

Authentication plus RBAC controls access.

ACCESS

4 Security

Zero Trust and layered controls limit blast radius.

PROTECT

THE POINT

Explain the mechanism, show the evidence, then state the trade-off.

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DOMAIN 3 · 30–35%

Describe Azure Management and Governance

03

Cost · governance · deployment · monitoring

Labs 12–16 · Blueprint effective 20 July 2026

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TECHNICAL MODEL

Azure cost factors become design inputs

1 Service and SKU

Compute size, storage tier and feature plan establish the rate card.

RATE

2 Region

Service availability, price and data residency vary by geography.

LOCATION

3 Usage

Runtime, transactions, retained capacity and network egress create quantity.

METER

4 Commitment

Reservations and savings plans trade flexibility for a lower effective rate.

TERM

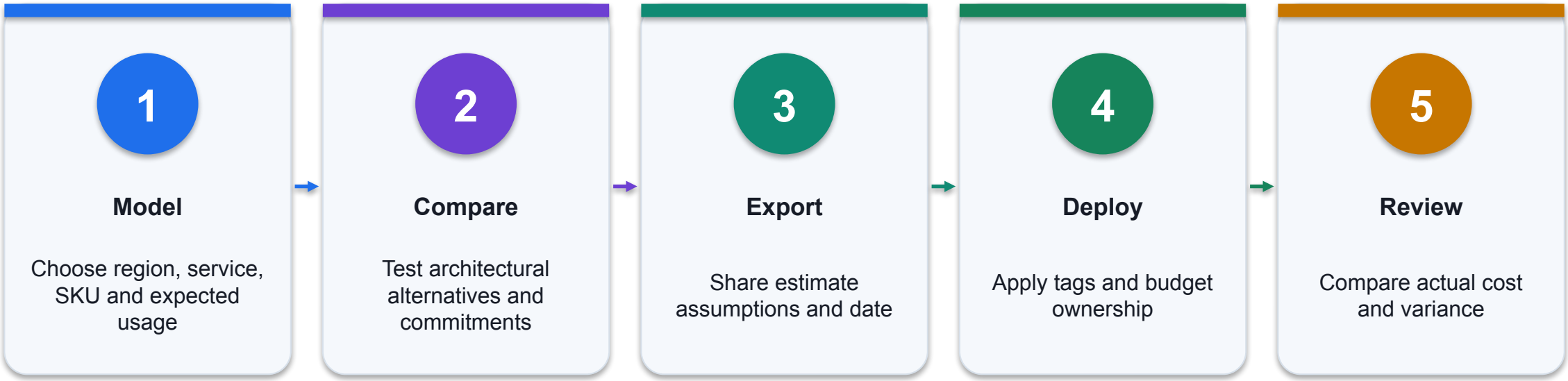
THE POINT

Cost is architecture feedback: surface the meter and owner before a workload reaches production.

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MECHANISM TRACE

Pricing Calculator to budget baseline



CONTROL RULE

An estimate is an assumption set; keep it with the design so variance can be explained.

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VISUAL SYSTEM MODEL

Governance and operations control loop



1 Prevent

Identity, RBAC and Policy constrain unsafe change.

2 Observe

Cost and monitoring show operational effect.

3 Respond

Alerts, diagnosis and remediation restore intended state.

Governance sets intended state; operations compares observed state and closes the gap.

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TECHNICAL MODEL

Tags are metadata, not security boundaries

1 Owner

Identify accountable team or service owner.

WHO

2 Environment

Separate prod, test and development reporting.

WHERE

3 Cost centre

Allocate spend to a business unit or project.

CHARGE

4 Automation

Select resources for policy, scripts and lifecycle tasks.

ACT

THE POINT

Tags support organisation and automation; they do not grant access and are not inherited automatically by every resource.

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DECISION MATRIX

Purview, Policy and locks

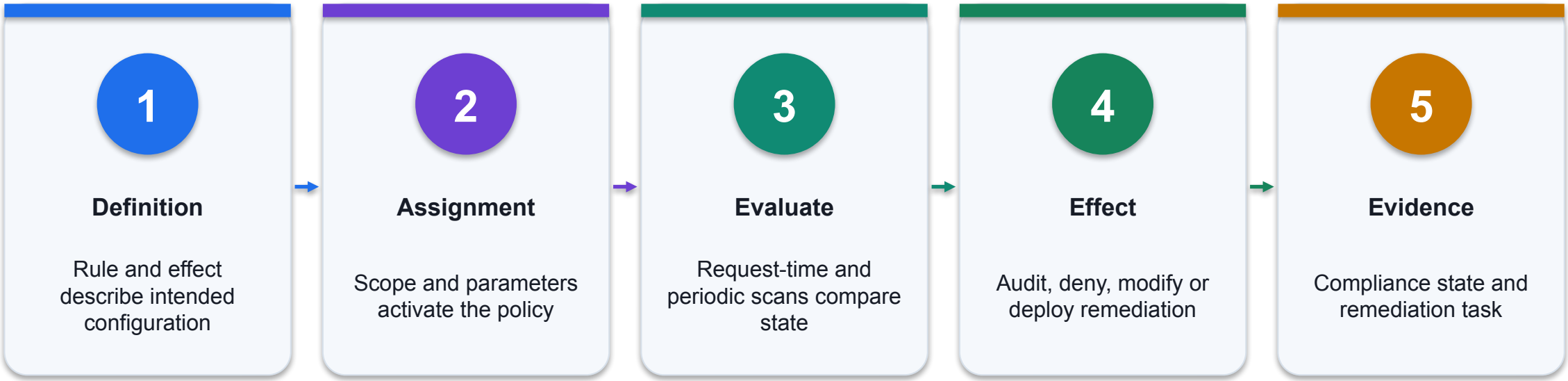
Control	Primary question	Mechanism	Evidence
Microsoft Purview	What data exists and how is it governed?	Catalog, classification, lineage, governance	Data map and classifications
Azure Policy	Is resource configuration allowed?	Audit, deny, modify, deployIfNotExists	Compliance state
Resource lock	Can authorised users change/delete this scope?	CanNotDelete or ReadOnly	Lock and failed operation

These controls complement one another: data governance, configuration governance and change protection.

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MECHANISM TRACE

Azure Policy evaluation path



CONTROL RULE

A policy definition does nothing until assigned at a scope with correct parameters and identity.

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DECISION MATRIX

Azure management interfaces

Interface	Strength	Repeatability	Evidence
Portal	Discovery and visual inspection	Manual unless exported	Activity/deployment logs
Cloud Shell	Browser-hosted CLI/PowerShell	Commands and scripts	Terminal output + logs
Azure CLI	Cross-platform automation	High	Command output + exit status
Azure PowerShell	Object pipeline and admin scripting	High	Objects + errors + logs

Use the portal to learn the resource model, then capture repeatable intent in commands or templates.

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TECHNICAL MODEL

Azure Arc extends the control plane

1 Servers

Project non-Azure Windows/Linux servers into Azure resource inventory.

HYBRID

2 Kubernetes

Attach clusters for inventory, policy and GitOps-enabled operations.

CLUSTER

3 Data services

Run selected Azure data capabilities on enabled infrastructure.

DATA

4 Governance

Apply tags, Policy and monitoring across distributed resources.

CONTROL

THE POINT

Arc does not move the workload into Azure; it extends Azure management and governance to the workload.

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TECHNICAL MODEL

Infrastructure as code contract

1 Declarative intent

Describe desired resources and properties instead of procedural clicks.

STATE

2 Parameters

Separate environment-specific values from reusable structure.

INPUT

3 Dependencies

Resource graph orders deployment and captures relationships.

GRAPH

4 Idempotence

Reapply the same template to converge on intended state.

REPEAT

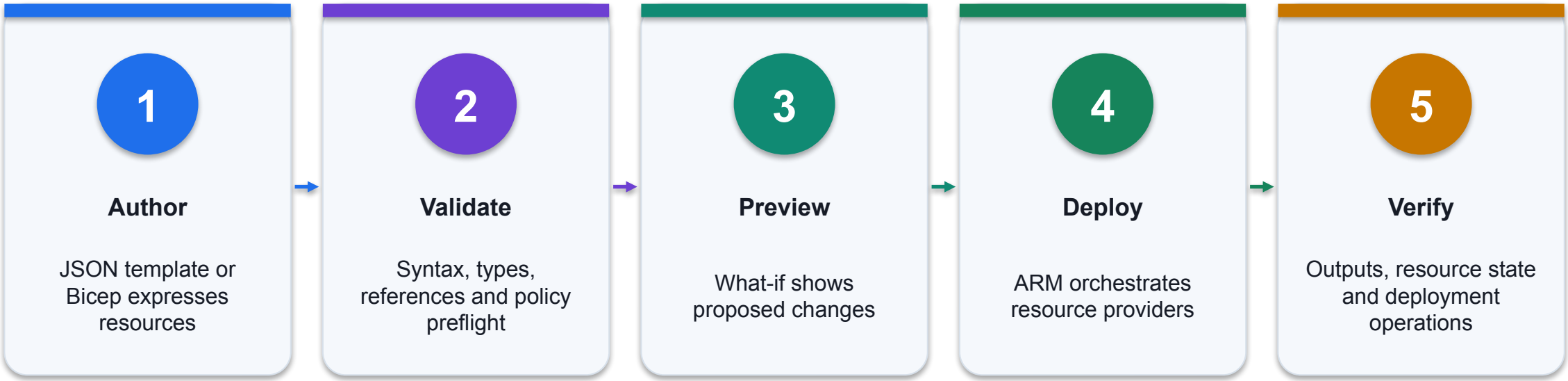
THE POINT

IaC improves repeatability only when versioning, review, validation and deployment evidence are part of the workflow.

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MECHANISM TRACE

ARM template deployment lifecycle



CONTROL RULE

Use what-if and deployment operations to understand change before and after execution.

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TECHNICAL MODEL

Azure Advisor recommendations

1 Reliability

Improve continuity and resilience configuration.

RESILIENCE

2 Security

Reduce exposure and improve posture.

PROTECT

3 Performance

Remove bottlenecks and tune service configuration.

SPEED

4 Cost

Right-size, shut down or commit where evidence supports it.

SAVE

THE POINT

Advisor provides recommendations; validate workload context and risk before applying a change.

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TECHNICAL MODEL

Service Health separates three views

1 Azure Status

Public global view of broad Azure service issues.

GLOBAL

2 Service Health

Personalised incidents, maintenance and advisories for your subscriptions.

YOU

3 Resource Health

Health of an individual resource and platform event history.

RESOURCE

4 Alert

Route relevant events through action groups to accountable responders.

ACT

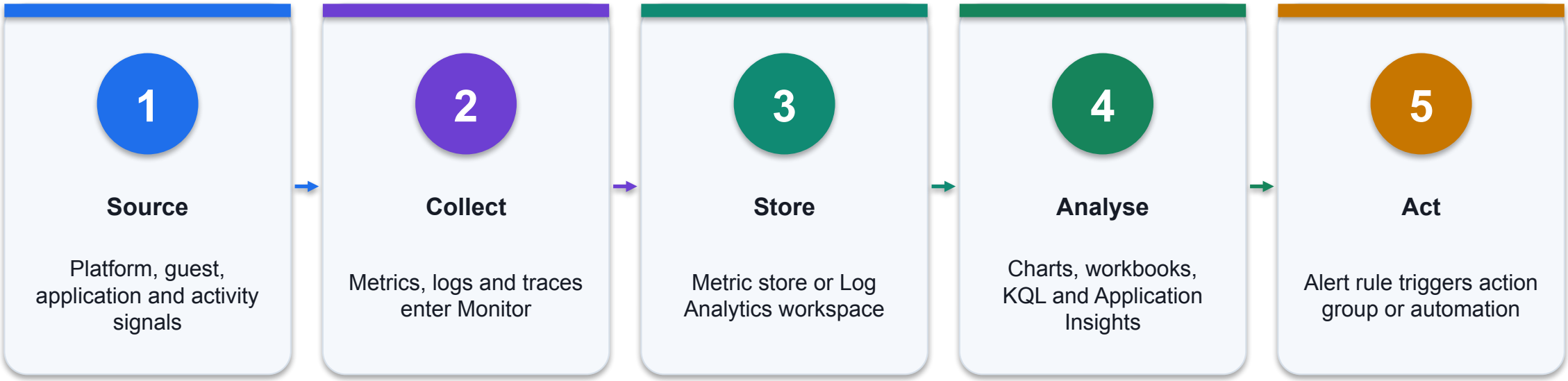
THE POINT

Start at the narrowest evidence: resource, subscription service event, then global status.

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MECHANISM TRACE

Azure Monitor telemetry pipeline



CONTROL RULE

An alert is only useful when its signal, threshold, owner and response action are testable.

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DECISION MATRIX

Metrics, logs and traces

Signal	Structure	Best for	Query
Metric	Numeric time series	Fast thresholding and trends	Metric explorer
Log	Timestamped records	Investigation and correlation	KQL in Log Analytics
Trace	Distributed request spans	Application dependency latency	Application Insights
Activity log	Control-plane events	Who changed what and when	Activity Log / export

Choose a signal that can prove the failure mode; collect everything is not an observability strategy.

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TECHNICAL MODEL

KQL query anatomy

1 Table

Select a record set such as AzureActivity or AppRequests.

FROM

2 Filter

where narrows time, resource, status or operation.

WHERE

3 Shape

project, extend and summarize transform records.

SELECT

4 Present

order by and render make the result interpretable.

OUTPUT

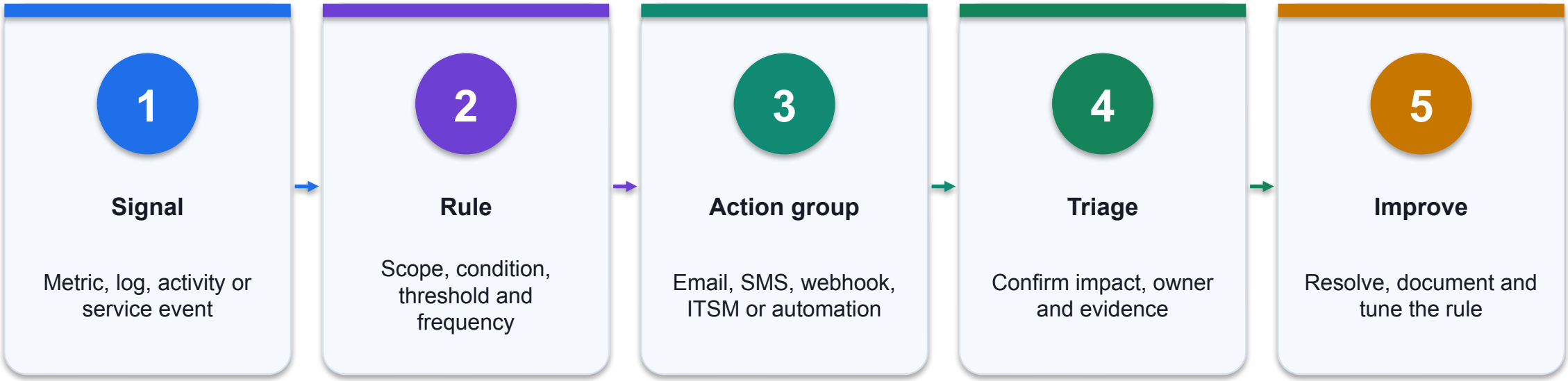
THE POINT

A useful query states the operational question, the time window and the evidence needed for a decision.

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MECHANISM TRACE

Alert-to-response control loop

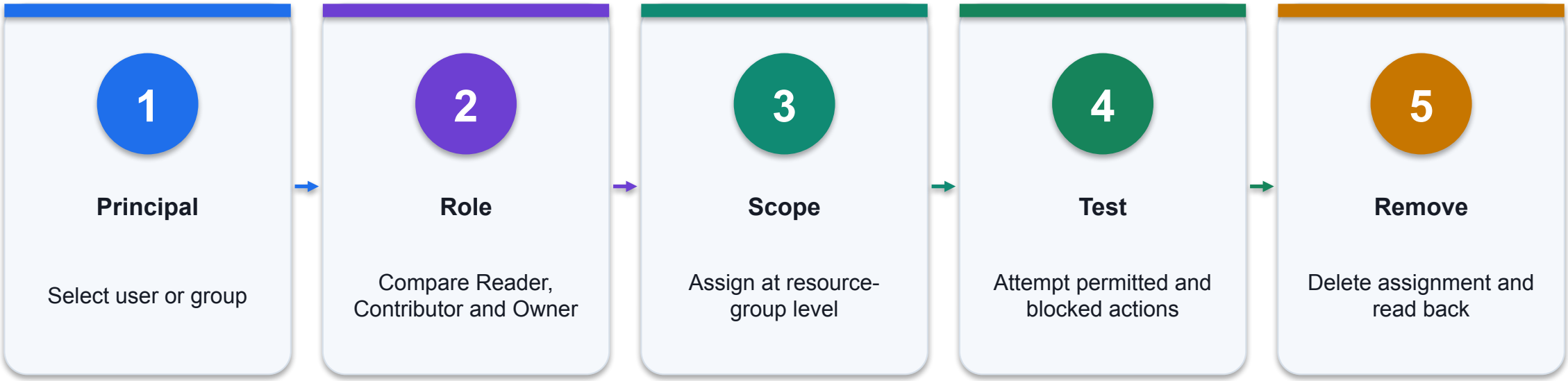


CONTROL RULE

Test alert routing before an incident and tune noisy rules using actual response outcomes.

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Lab 12 · Manage Access with Azure RBAC



CONTROL RULE

Deliverable: Reader can view rg-az900-labs but cannot create or delete resources; assignment is removed after the test. Full click-by-click procedure: Learner Guide and labs/lab-12-*.md.

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Lab 12 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Reader can view rg-az900-labs but cannot create or delete resources; assignment is removed after the test.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

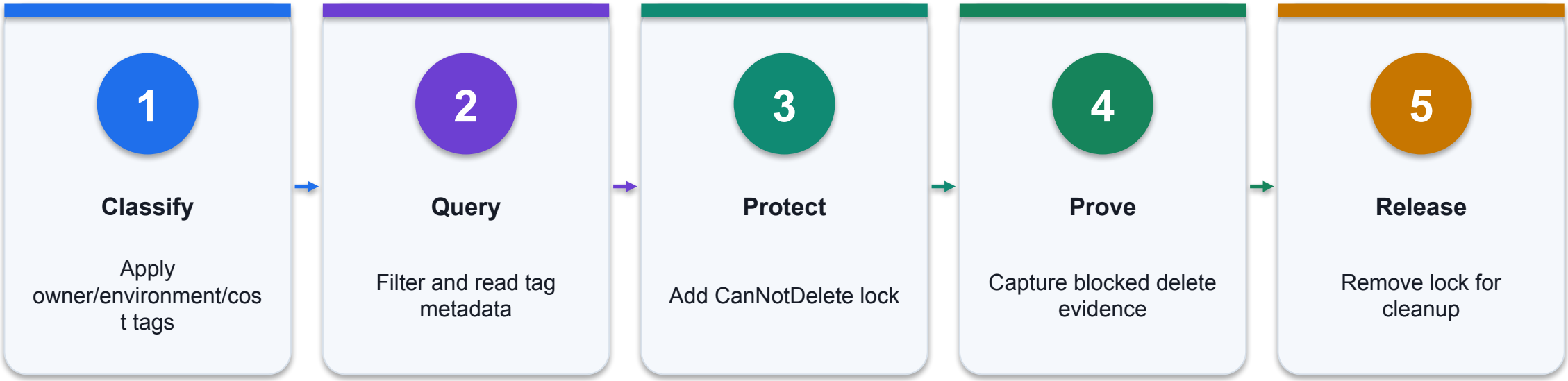
Unexpected access: inspect inherited assignments and use Check access for the exact principal.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 13 · Apply Resource Tags and Resource Locks



CONTROL RULE

Deliverable: Tags read back at intended scopes and the lock blocks deletion until explicitly removed. Full click-by-click procedure: Learner Guide and labs/lab-13-*.md.

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Lab 13 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Tags read back at intended scopes and the lock blocks deletion until explicitly removed.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

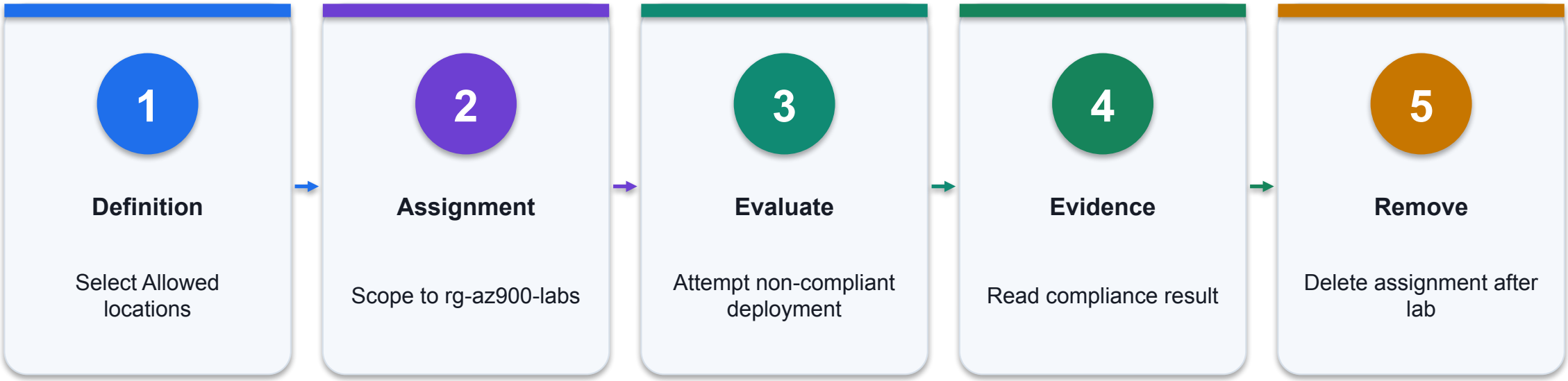
Cleanup blocked: enumerate locks at resource, group and subscription scopes.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 14 · Enforce Standards with Azure Policy



CONTROL RULE

Deliverable: A disallowed-region deployment is denied and the assignment/compliance state is visible. Full click-by-click procedure: Learner Guide and labs/lab-14-*.md.

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Lab 14 · Verification and diagnosis

ACCEPTANCE EVIDENCE

A disallowed-region deployment is denied and the assignment/compliance state is visible.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

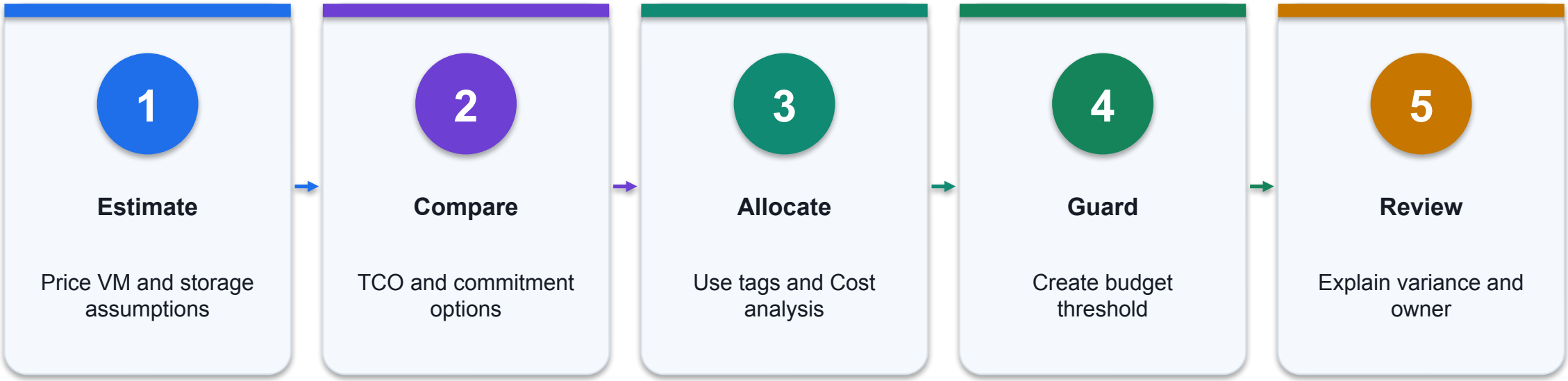
No denial: confirm assignment scope, parameters, enforcement mode and propagation time.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 15 · Estimate and Control Costs



CONTROL RULE

Deliverable: Estimate assumptions are saved and a budget exists with an accountable notification route. Full click-by-click procedure: Learner Guide and labs/lab-15-*.md.

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Lab 15 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Estimate assumptions are saved and a budget exists with an accountable notification route.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

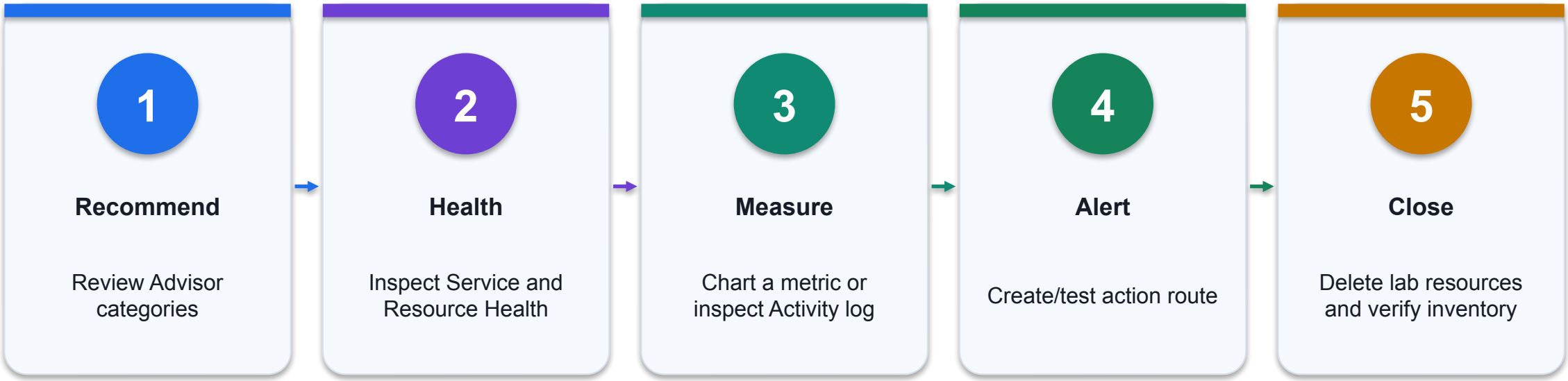
No cost data: remember reporting latency and verify subscription, date range and permissions.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Lab 16 · Monitor Your Environment and Complete Cleanup



CONTROL RULE

Deliverable: Monitoring evidence is captured and final resource inventory contains no unintended billable lab resources. Full click-by-click procedure: Learner Guide and labs/lab-16-*.md.

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Lab 16 · Verification and diagnosis

ACCEPTANCE EVIDENCE

Monitoring evidence is captured and final resource inventory contains no unintended billable lab resources.

Capture the named resource state, command output or service result. Evidence must match the intended subscription and scope.

LIKELY FAILURE

Residual resources remain: query by resource group/tag and inspect locks, dependencies and hidden child resources.

DIAGNOSTIC ORDER

1 · Identity and subscription
2 · Resource and configuration state
3 · Network path and service logs
4 · Cleanup and cost exposure

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Domain 3 · Decision recap

1 Govern

Policy, locks and metadata express intended state.

INTENT

2 Deploy

IaC makes change reviewable and repeatable.

CHANGE

3 Observe

Metrics, logs and health provide operational evidence.

SIGNAL

4 Improve

Cost and alert feedback close the control loop.

ACT

THE POINT

Explain the mechanism, show the evidence, then state the trade-off.

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COURSE RECAP

What you can now explain and verify

1 Cloud decisions

Choose models and service types by responsibility and control.

CONCEPT

2 Azure architecture

Trace regions, resource hierarchy and service contracts.

ARCH

3 Governance

Connect identity, RBAC, Policy, cost and monitoring evidence.

CONTROL

4 Hands-on evidence

Create, verify, diagnose and clean up sixteen Azure lab scenarios.

PROOF

THE POINT

Use the official July 2026 skills outline as the final revision checklist.

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LEARNER ACCESS

Practice and Exam Readiness

learn.microsoft.com/.../az-900/practice-assessment

Official Microsoft Learn AZ-900 practice assessment

1. **Open the official practice assessment**
2. **Attempt a timed first pass**
3. **Review explanations by domain**
4. **Return to weak mechanisms and labs**

1 **Current check**

Tertiary Exams did not list an AZ-900 bundle at release.

2 **Official source**

Use Microsoft Learn for the matching practice assessment.

3 **No dumps**

Practise concepts and reasoning, not memorised items.

4 **Retake**

Track domain improvement before exam day.

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Assessment reminder

1 WA

1 hour · open-ended K1–K9 · open book.

4:00–5:00

2 PP

1 hour · 3 lab-aligned tasks · open book.

5:00–6:00

3 Evidence

Submit required screenshots, outputs and explanations.

VALID

4 Privacy

Do not share candidate papers or answer keys outside authorised channels.

PRIVATE

THE POINT

Take assessment digital attendance first, then submit completed candidate papers on the LMS.

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Assessment flow · final check



CONTROL RULE

Do not leave before the assessor confirms submission and sign-off.

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Digital Attendance · Mandatory

https://lms-tms.tertiaryinfotech.com/

Complete the assessment checkpoint

1. **Open the LMS attendance tile**
2. **Scan the current TRAQOM QR code**
3. **Submit the attendance response**
4. **Show confirmation if requested**

1 **System**

TRAQOM is SSG digital attendance.

2 **Timing**

Complete before assessment starts.

3 **Record**

Trainer checks the captured status.

4 **Support**

Raise access issues immediately.

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THANK YOU

Microsoft Azure Fundamentals (AZ-900)

Explain the mechanism · verify the evidence · control the cost

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